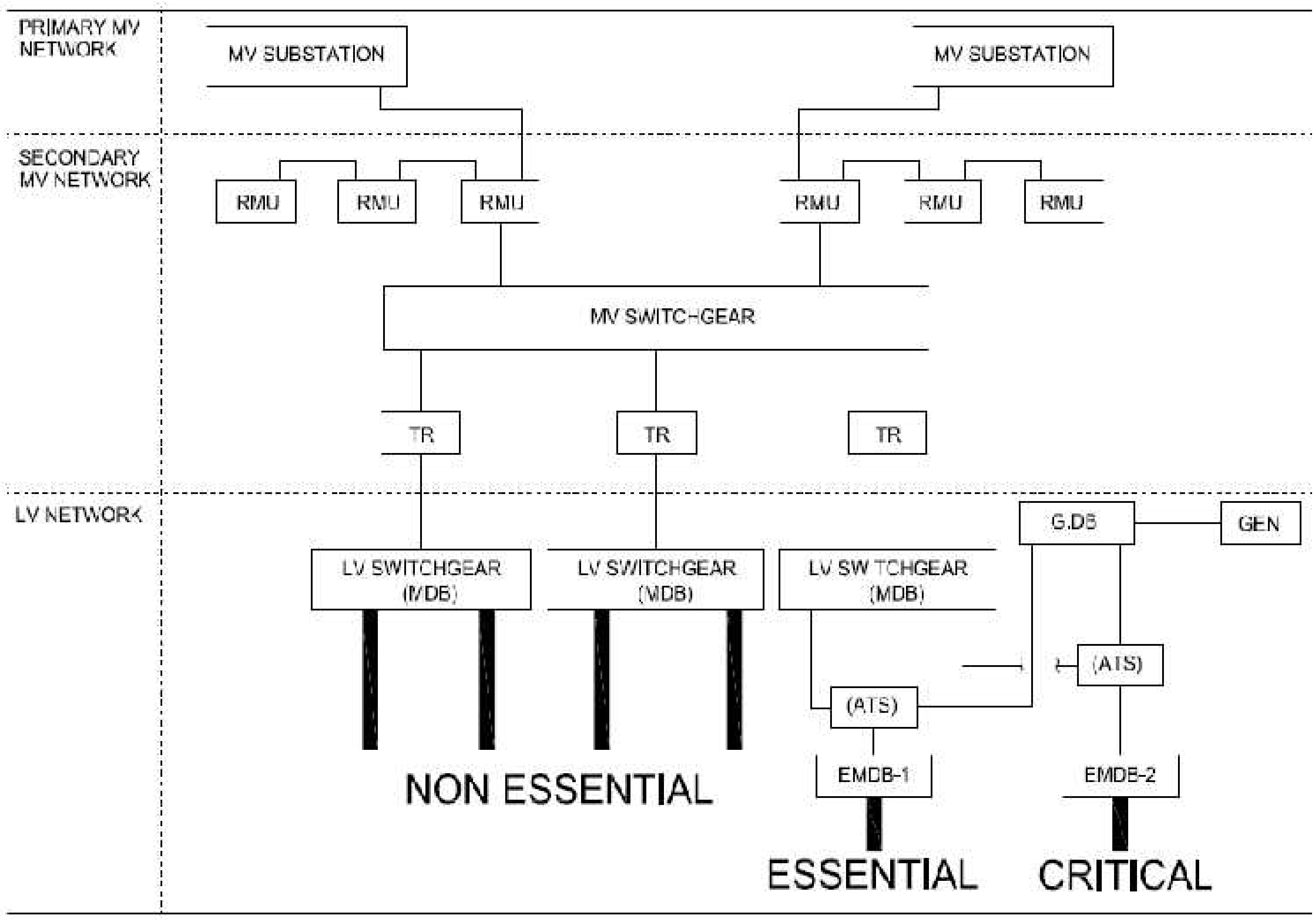
MV switchgear IEC 62271-200

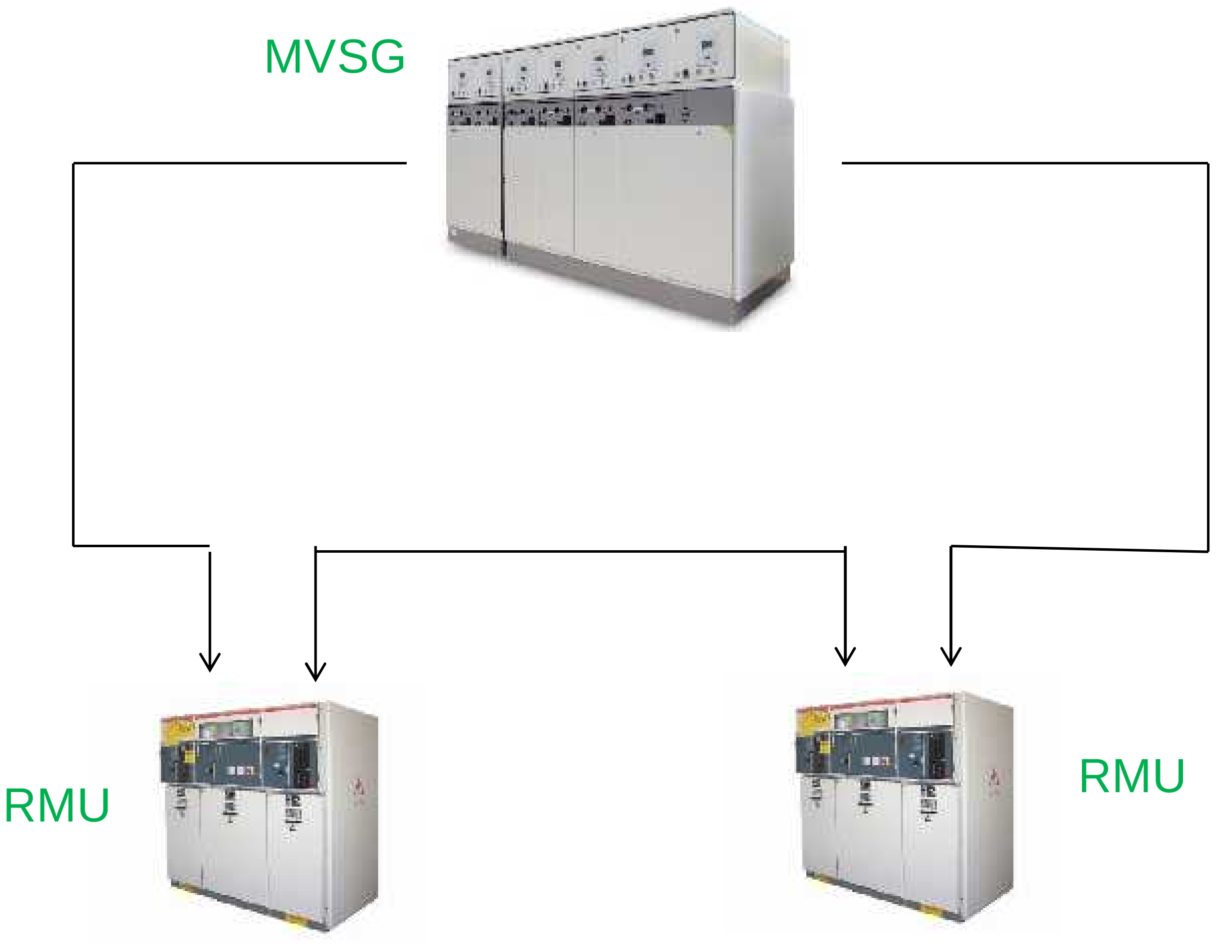
Introduction

Distribution Network Configuration



Introduction

Distribution Network Configuration



Introduction

The main reasons of the new standard

- The old standard was 13 years old and was based on technologies of more than 30 years.

- The new standard:
  - better cover of solutions available on the market
  - improves wording of basic equipment
  - value the experience gained on internal arc
  - is more customer value oriented

Introduction

The main changes in the new IEC 62271-200 standard

- The standard IEC 62271-200:
  - replaces the former IEC 60298
  - Has been published in december 2003
- The standard IEC 62271-200 specifies requirements:
  - for factory-assembled metal-enclosed switchgear and controlgear
  - for alternating current of rated voltages above 1kV and up to 52kV
  - for indoor or outdoor installation
  - for services frequencies up to and including 60Hz
  - AIS - GIS - OIS
  - P < 300kPa (3 bar)
- The standard partially includes the requirements of the IEC 60694 “common clauses for high voltage switchgear and controlgear”
- It adds specific requirements for metal-enclosed switchgear

Introduction

The main changes in the new IEC 62271-200 standard

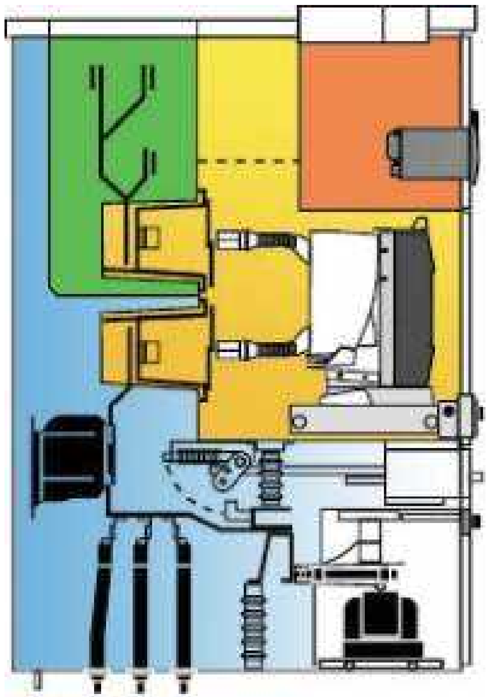
- Partitioning:
  - the manufacturer defines the number and content of compartments
  - each compartment is described as fixed or withdrawable and accessibility
- Nature of barrier:
  - between live parts and opened accessible compartment
- Service continuity:
  - the main customer value is the continuity of service provided by the switchgear under conditions when an accessible compartment is open
- Functional unit of a switchboard:
  - is based on functionality rather than on design and construction features
- Internal arc fault:
  - becomes an optional rated performance through a type test completely defined in the standard

Partitioning & Nature of barrier

Compartments: description of design and function

- The manufacturer defines the number & content of compartments

- Each compartment is described as:
  - design:
    - fixed
    - removable
    - withdrawable
    - extractible
  - accessibility:
    - interlocked-controlled
    - procedure based
    - tools based
    - none-accessible



Partitioning & Nature of barrier

The nature of the partitioning class

Switchgear classification with regard to the nature of the barrier between live parts and opened accessible compartment.	Features
PM	Metal shutters and partitions between live parts and open compartment (metal-enclosed condition maintained).
PI	Insulation - covered discontinuity in the metal partitions / shutters between live parts and open compartment.

Service continuity

The Loss of service continuity classes

- The LSC classes are describing functionality and design related features.
- The LSC classification is not a ranking of service continuity.

Switchgear categories with regard to loss of service continuity when opening compartments (other than busbar, in single busbar designs)		Features
LSC1		Other functionnal units or some of them shall be disconnected
LSC2	LSC2A	All other functional units can be energized
	LSC2B	All other functional units and all cable compartments can be energized

Service continuity

Loss of Service Continuity Class (LSC)

- Architecture based on "safe compartment access"

Several levels of service continuity during maintenance

LSC 2A

- Safe access to compartment
- With power flow in busbar and the other units
- MV Cables must be earthed

Maintainability of one functional unit allowing normal service of the remaining units of the switchboard (busbar in a separate compartment)

LSC 2B

- Safe access to compartment
- With power flow in busbar and the other units
- MV Cable in separate compartment
- Cable of unit under maintenance can remain energised

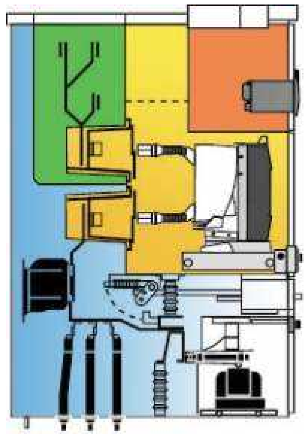
Maintainability of defined parts with no need of cable disconnection (separate cable compartment)

LSC 1

- Metal enclosed not of LSC2 class

Internal arc fault & classification

The test procedure is clearly defined and mandatory



- The switchboard installation condition is simulated (floor, ceiling, operator)
- 2 or 3 functional units connected together are tested
- Each compartment containing a main circuit component is subjected to testing:
  - busbar compartment
  - circuit-breaker compartment
  - cable connection compartment
- Arc ignition location points are specified
- Feeding directions specified according to the compartment under test

Internal arc fault & classification

MCset meets the 5 criteria of IEC 62271-200

- Criterion 1: correctly secure doors and covers do not open
- Criterion 2: no fragmentation of the enclosure occurs within the time specified for the test
- Criterion 3: arcing does not cause holes in the accessible sides up to a height of 2m
- Criterion 4: indicators do not ignite due to the effect of the hot gases
- Criterion 5: the enclosure remains connected to its earthing point

Internal arc fault & classification

Types of accessibility

Metal-enclosed switchgear and controlgear, except pole mounted:

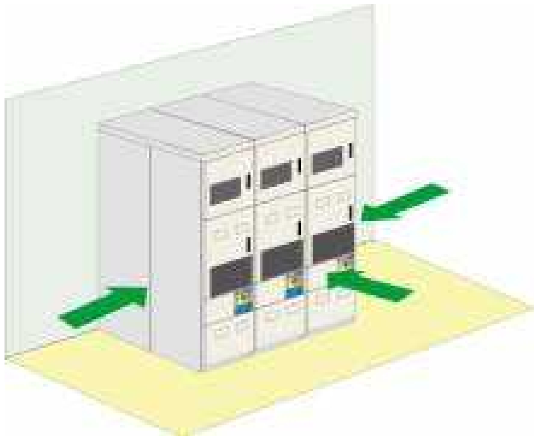
Accessibility Type A		Restricted to authorized personnel only.
Accessibility Type B		Unrestricted accessibility, including that of the general public.
According the accessibilities A and B type, the following code shall be used:	F: for Front side L: for Lateral side R: for Rear side	

Pole-mounted metal-enclosed switchgear and controlgear:

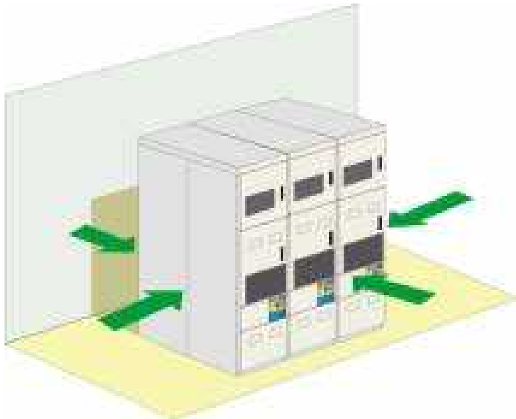
Accessibility Type C		Accessibility restricted by installation out of reach.

Internal arc fault & classification

- Examples of accessibility on the various sides of the switchgear is codified Annex A.2



Switchboard against a wall. AFL anti-arcing protection



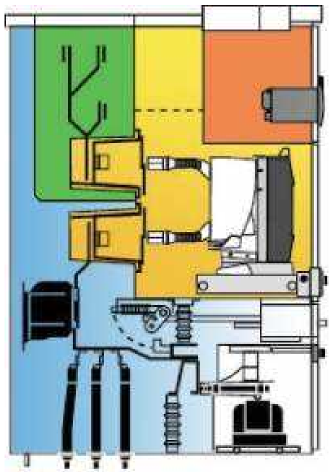
Switchboard in the middle of the room. AFLR anti-arcing protection (acces around the back of the switchboard)

Switchboard in a room with a ceiling height of less than 4 meter

Conclusion

MCset service continuity & classification

List of compartments	Busbar	Cables	Circuit-Breaker	VTs
Design	Fixed	Fixed	Withdrawable	Withdrawable
Accessibility	With Tools	Interlocked	Interlocked	Interlocked
Partitioning	Metallic everywhere			



Classification for MCset:
Service continuity: LSC2B
Partitioning: PM
Accessibility: Type A
Code: FL - FLR

Conclusion

The new standard is based on functionality rather than on design and construction features:

- Top down approach:
  - functionality (type of functions, fixed or removable, architecture and compartments, need for maintenance)
  - start from the real needs in terms of condition for accessibility and continuity of service.
- Consider all the technologies available
- Internal arc withstand is a completely defined optional feature

RMU :

	RM6	SM6
TYPE	GIS	AIS
Partitioning class	PM	PI
IP	67	3X
Optimized dimensions	yes	no
Maintenance free	Yes	no
Width	500 mm	500 mm
Depth	710 mm	930 mm
Hight	1142 mm	1600 mm + 400 mm for exhaust

RM6

SM6